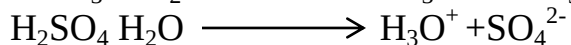
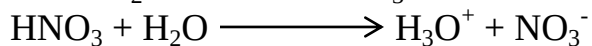
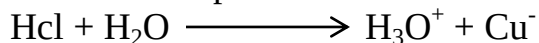


ACID, BASE AND SALT

An acid is a substance which in an aqueous solution, produces hydroxonium (H_3O^+) as the only positive ion, by bronsted and lower definition, an acid is a proton donor. Examples:



The above three common acid are the most common mineral (inorganic) acids in the laboratory. There are pronounced as follows:

HCl – Hydrochloric acid

HNO_3 – Trioxonitrate (V) acid

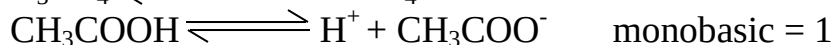
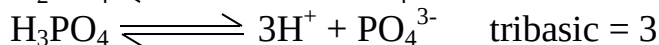
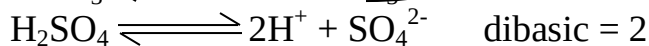
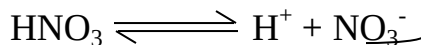
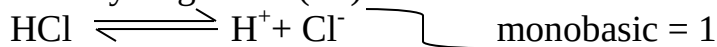
H_2SO_4 – Tetraoxosulphate (VI) acid.

Many acids occur naturally

Natural acid	source
Butanoic acid	human sweat
Citric acid	lemons
Ethanoic acid	venigar and palmwine
Lactic acid	milk
Tartaric acid	grapes

BASICITY OF AN ACID

The basicity of an acid refers to the number of hydroxonium ions which one molecule of the acid is capable of producing in aqueous solution. OR the number of replaced hydrogen ion (H^+) in one molecule of the acid. Examples:



NB: Basicity of an acid is not the number of hydrogen atoms contained in one molecule of the acid.

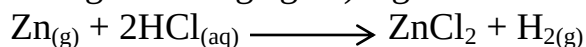
Ethanoic acid CH_3COOH has four hydrogen atoms per molecule but only one is replaceable by a metal, hence, it is monobasic acid.

PHYSICAL PROPERTIES OF ACIDS

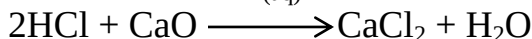
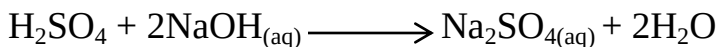
1. Ionic form of strong acid are corrosive (can burn).
2. Dilute acids have sour taste
3. Acids turn blue litmus paper to red.

CHEMICAL PROPERTIES OF ACIDS

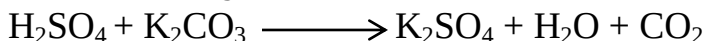
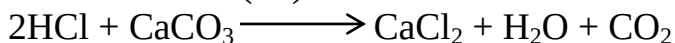
1. Reaction with metals: More electropositive metals than hydrogen displace/liberate hydrogen gas from an acid. An exception is HNO_3 (it is a strong oxidizing agent) e.g.



2. Reaction with bases and alkalis: Acids react with insoluble bases and alkalis to form salt and water only. The reaction between an acid and alkali is called neutralization.

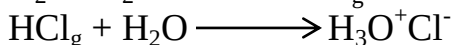
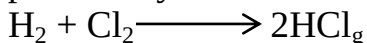


3. Reaction with trioxocarbonate (IV): Acids react with trioxocarbonate (IV) to liberate carbon (IV) oxide.

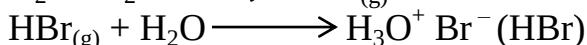
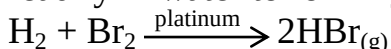


PREPARATION OF ACIDS

1. By direct combustion of constituent elements using a suitable catalyst e.g. hydrogen gas combine with chlorine gas in the presence of activated charcoal as catalyst to give hydrogen chloride gas which dissolve in water to produce hydrochloric acid.

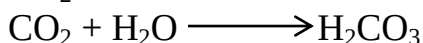


- b. Hydrogen gas also combines with bromine vapour in the presence of platinum as catalyst to produce hydrogen bromide gas which dissolves readily in water to form hydrobromic acid.

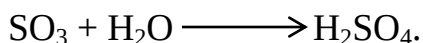


2. By the reaction between an acid anhydride and water

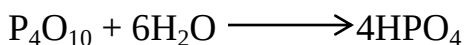
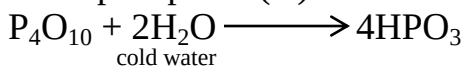
- a. CO_2 dissolves in water to form a weak acid (trioxocarbonate (IV) acid)



- b. Sulphur (IV) dissolves in water to form a strong acid, tetraoxosulphate VI acid.

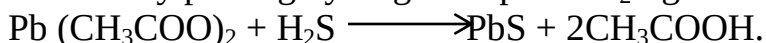


- c. Phosphorous (V) oxide combines readily with cold or hot water to form trioxophosphate (V) and tetraoxosulphate (V) acid respectively.



3. By displacing a weaker or more volatile acid from its salt by a stronger or less volatile acid.

4. By precipitation: Precipitation of an insoluble sulphide from its metallic salt solution by passing hydrogen sulphide H_2S gas into it.



USES OF ACID

1. In the manufacturing of fertilizers, dyes, detergents, explosives, paints, soaps, textiles, drugs, plastics and other chemicals.
2. In dissolving metals to remove rust.
3. In the production of salts, soft drinks and baking powder.
4. As food preservatives and mud antiseptic or germicide.

5. In oil refineries.
6. As electrolyte in lead acid accumulation.
7. As a dyeing, dehydrating and oxidizing agent.
8. Wasp stings are alkaline and can be neutralized with ethanoic acid (e.g. vinegar).

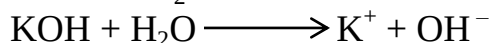
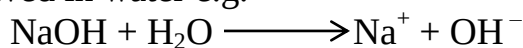
BASES AND ALKALIS

Bronsted and Lowry defined base as proton acceptor.

A base is a substance which neutralizes an acid to yield a salt and water only.

Oxides and hydroxides of metals are bases.

Examples: Na_2O (sodium oxide), K_2O (potassium oxide), MgO (magnesium oxide). These bases are insoluble in water. A soluble hydroxide is known as alkalis. An alkali is a compound which produces hydroxyl ion OH^- when dissolved in water e.g.



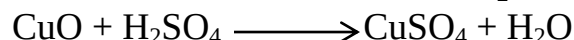
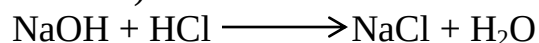
Sodium and potassium hydroxides are strongly alkalis because they ionise completely in aqueous solution. $\text{Ca}(\text{OH})_2$ is a weak alkali. NH_4OH is also a weak alkali.

PHYSICAL PROPERTIES OF BASES/ALKALIS

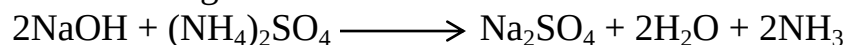
1. Alkalis have a bitter taste.
2. Alkalis turn red litmus paper blue
3. Alkalis have soapy feel
4. Common forms of the two caustic alkalis NaOH and KOH have a powerful corrosive action.

CHEMICAL PROPERTIES OF BASES/ALKALIS

1. Reaction with acids: All bases react with acids to form salts and water only (neutralization).

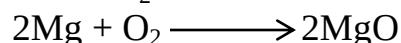
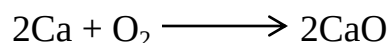


2. Reaction with ammonium salts: All alkalis react with ammonium salts to generate ammoniac gas.

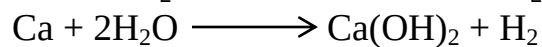
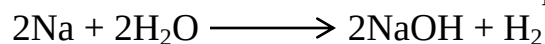


PREPARATION OF BASES/ALKALIS

1. Most metals combine directly with oxygen when heated to form basic oxides.



2. Some metals react with water or steam to produce alkalis.



3. Decomposition of trioxocarbonate (IV).

